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Normalized Dataset for Sanskrit Word Segmentation and Morphological Parsing

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Abstract

Sanskrit processing has seen a surge in the use of data-driven approaches over the past decade. Various tasks such as segmentation, morphological parsing, and dependency analysis have been tackled through the development of state-of-the-art models despite working with relatively limited datasets compared to other languages. However, a significant challenge lies in the availability of annotated datasets that are lexically, morphologically, syntactically, and semantically tagged. While syntactic and semantic tags are preferable for later stages of processing such as sentential parsing and disambiguation, lexical and morphological tags are crucial for low-level tasks of word segmentation and morphological parsing. The Digital Corpus of Sanskrit (DCS) is one notable effort that hosts over 650,000 lexically and morphologically tagged sentences from around 250 texts but also comes with its limitations at different levels of a sentence like chunk, segment, stem and morphological analysis. To overcome these limitations is to look at alternatives such as Sanskrit Heritage Segmenter (SH) and Samsaadhanii tools, that provide information complementing DCS' data. This work focuses on enriching the DCS dataset by incorporating analyses from SH, thereby creating a dataset that is rich in lexical and morphological information from both DCS and SH. Furthermore, this work also discusses the impact of such datasets on the performances of existing segmenters, specifically the Sanskrit Heritage Segmenter.

Keywords: Word Segmentation, Morphological Parsing, Datasets, Sanskrit Computational Linguistics

1 Introduction

Word Segmentation is the first step towards analysing sentences in any language. The subsequent tasks of morphological parsing, POS tagging, dependency parsing, sense disambiguation and discourse analysis, all require in first place, the individual words of a sentence. This is a low-level task of Natural Language Processing where the unsegmented sentence is split into lexically and morphologically valid words. Languages like English and German have explicit boundaries like space which indicate the split locations. Certain other languages like Sanskrit, Chinese and Japanese have a tendency to join the words making the process of segmentation challenging. Some of these are concatenation of words (as in Chinese) while some have a euphonic transformation (as in Sanskrit) at the word boundaries.

In order to arrive at the individual words in any language, the primary requirement is the notion of a word. The definition of a word varies according to the language and hence language-specific word segmentation tasks are near-perfect for some languages (Shao et al, 2018). In Sanskrit, a word (*padam*)¹ is either a noun form or a verb form.² Segmenting a Sanskrit sentence into such individual words requires lexical and morphological validation of the words thus formed. In addition to that, there is a tendency to write Sanskrit texts continuously without any breaks, with the letters at the word boundaries undergoing euphonic transformations termed as *sandhi*. Pāṇini's *Aṣṭādhyāyī*, the well celebrated Sanskrit grammar, records formation of all the possible *sandhis*. There are two types of *sandhis*: internal and external. Internal *sandhis* correspond to the transformation that occurs within a word, predominantly occurring during the formation or generation of a word. External *sandhis* occur across words and at the word boundaries. The term *sandhi* in this paper refers to external *sandhi* by default, unless specified otherwise. An example of *sandhi* occurring in a sentence can be observed in the unsegmented-segmented pair of the following verse (*Bhagavad Gītā* 1.1):

Unsegmented

*dharmakṣetre kurukṣetre samavetā yuyutsavaḥ
māmakāḥ pāṇḍavāścaiva kimakurvata sañjaya*

Segmented

*dharm-a-kṣetre kuru-kṣetre samavetāḥ yuyutsavaḥ
māmakāḥ pāṇḍavāḥ ca eva kim akurvata sañjaya*

We notice the following euphonic transformations that occur in the verse:

- *samavetā yuyutsavaḥ* → *samavetāḥ yuyutsavaḥ*³
- *pāṇḍavāścaiva* → *pāṇḍavāḥ ca eva*⁴
- *kimakurvata* → *kim akurvata*

¹ *suptiñantaṃ padam* (*Aṣṭādhyāyī* 1.4.14)

² The nominal stems (*prātipadikas*) are inflected with the nominal suffixes (*sup*) to produce the nominal forms (*subantas*) and the verbal roots are inflected with the verbal suffixes (*tiñ*) to produce the verbal forms (*tiñantas*).

³ *visarga-lopa-sandhiḥ*

⁴ *ścutva-sandhiḥ* and *vṛddhi-sandhiḥ* respectively

- *dharmakṣetre* → *dharmak-ṣetre*, “-” indicating that it is a compound word. Similarly *kurukṣetre*.

Compounds are special constructions of words where two words combine to form a new word with a different meaning, either modifying the first or the second word, or referring to something which is totally different from the referents of its constituents. In this case, *dharmā* (duty) and *ṣetre* (in the field / place), combine to form *dharmakṣetre* (in the place of duty) and is derived as *dharmasya* (of duty) *ṣetre* (in the place).

Segmenting a Sanskrit sentence thus involves three tasks:

1. resolving *sandhis*,
2. identifying valid words, and
3. deciding whether the split words constitute a compound or are standalone words

The rules of *Sandhi* along with a morphological analyser to validate the segments do help in the segmentation process. In spite of such validation, there is always a possibility of non-determinism in segmentation, even in segmenting a word as simple as *rāmālayaḥ*, as shown below:

- (a) If we consider only the surface forms, then there are 7 possible solutions (table 1). Of these, the first two analyses are compounds and the remaining are standalone words. Only the first two are grammatically and semantically meaningful. The remaining five are either grammatically or meaningwise incompatible.
- (b) The task of segmentation further becomes complicated, if one also selects the segments not on the basis of forms but on the basis of their morphological analyses. The two factors homonymy and syncretism thus bring in non-determinism, as illustrated below. Let us consider the words *rāmaḥ* and *vanam*.

- **Homonymy** → depending on the context, *rāmaḥ* can either be a noun (masculine singular nominative form of the stem *rāma* (indicating a person named Rāma)) or a verb (present active first person plural form of the root *rā* (to give)).
- **Syncretism** → *vanam* can either be in the nominative or in the accusative case for the stem *vana* (a forest).

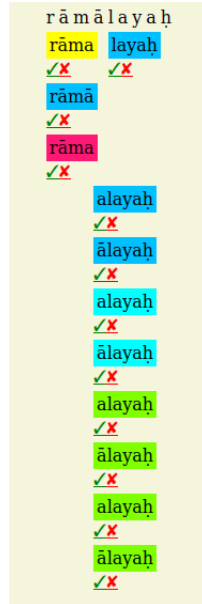
In the example *rāmālayaḥ*, if we consider the surface forms along with the morphological and lexical analysis, there are 17 possible solutions (figure 1). The colors correspond to different morphological analyses indicating different parts of speech. Among these again only 8 are grammatically and semantically compatible splits.

With these challenges, the task of Sanskrit word segmentation has seen several approaches ranging from rule-based methodologies to finite state automata and towards machine learning methods as well. The rules from *Aṣṭādhyāyī* alone are insufficient for the word segmentation task due to the indeterminacy in resolving *sandhi*. A morphological analyser too validates all possible analyses to true, since it looks at a single word at a time and does not take into account the context. This results into a large number of solutions. There

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Table 1 Possible Segmentations the word *rāmālayaḥ* (word-forms)

Solutions of <i>rāmālayaḥ</i>	Meaning
1 <i>rāma-ālayaḥ</i>	<i>rāma</i> -(Lord Rama's) <i>ālayaḥ</i> (abode)
2 <i>rāma-alayaḥ</i>	<i>rāma</i> -(Lord Rama's) <i>alayaḥ</i> (scorpions / bees) <i>rāma</i> -(Lord Rama's) <i>alayaḥ</i> (non-destruction / non-dissolution)
3 <i>rāma ālayaḥ</i>	<i>rāma</i> ((we) give / bestow) <i>ālayaḥ</i> (abode)
4 <i>rāma alayaḥ</i>	<i>rāma</i> ((we) give / bestow) <i>alayaḥ</i> (scorpions / bees) <i>rāma</i> - ((we) give / bestow), <i>alayaḥ</i> (non-destruction / non-dissolution)
5 <i>rāmā layaḥ</i>	<i>rāmā</i> (pleasant / charming) <i>layaḥ</i> (absorption in / clinging to)
6 <i>rāmā ālayaḥ</i>	<i>rāmā</i> (pleasant / charming) <i>ālayaḥ</i> (abode)
7 <i>rāmā alayaḥ</i>	<i>rāmā</i> (pleasant / charming) <i>alayaḥ</i> (scorpions / bees) <i>rāmā</i> (pleasant / charming) <i>alayaḥ</i> (non-destruction / non-dissolution)

Fig. 1 Possible Segmentations of the word *rāmālayaḥ* (word-forms and morphological analyses)

have been efforts towards building a segmenter that produces all possible analysis, like Sanskrit Heritage Segmenter (SH).⁵ Choosing the intended segmentation requires additional information like the context. Machine learning approaches do provide a way out but the unavailability of sufficient annotated

⁵<https://www.sanskrit.inria.fr/>

dataset stands as an obstacle. There have been efforts towards building an annotated dataset, like the Digital Corpus of Sanskrit (DCS)⁶ (Hellwig, 2010–2021) and the dataset for word segmentation (Krishna et al, 2017) (referred to as SIGHUM),⁷ which are commendable but these datasets also come with limitations of their own.

This paper focuses on building a larger dataset with more linguistic features, using DCS as the base dataset, SH as the base segmenter and partially the methodology incorporated by Krishna et al (2017). SH produces all possible segments with the lexical and morphological analyses for each of the segments in the form of a graphical interface where the users can choose or reject a segment (as shown in figure 1). DCS has more than 650,000 sentences annotated with the information regarding lemma and morphological analysis. Krishna et al (2017) aligned the ground truth values of the DCS analyses with the possible analyses from SH, producing a set of 119,000 sentences with the combined analysis of both DCS and SH. This alignment process was modified by Krishnan et al (2023) for all the sentences from DCS which resulted into a dataset of 130,000 sentences. It is now extended further in this paper, using the latest DCS annotations, with modifications to the alignment process and creation of a larger dataset. The challenges observed during the creation of this dataset are also recorded along with comparisons on the design decisions of DCS and SH. The resultant dataset is then used for ranking the solutions of SH and the performance comparisons of the updated SH (referred further as SH-Ranking) with the available word segmenters are also recorded.

The paper is structured as follows. Section 2 gives a brief note on the recent developments in word segmentation for different languages and Sanskrit. Section 3 describes the SH along with its advantages and limitations. Section 4 enlists the available datasets and gives an account of each of these. Section 5 describes the differences between the design decisions of latest DCS representation and SH’s analysis. The differences are recorded with respect to chunk, stem, morphological analysis and compounds. The current process of the alignment and dataset creation is described in section 6. Section 7 details out the developments to SH with the help of the alignment process. Section 8 records the observations on the performance of SH-Ranking and recent segmenter models. Finally, inferences and discussions are in section 9.

2 Recent Developments in Word Segmentation

2.1 In other languages

Looking at word segmentation in languages with concatenation (Thai, Chinese and Japanese), there are two basic issues: word-boundary disambiguation and unknown word identification (Gao et al, 2005) where disambiguation happens between compound forms and individual word forms. In Thai, possible word

⁶<http://www.sanskrit-linguistics.org/dcs/>

⁷<https://zenodo.org/record/803508#.YRdZ43UzaXJ>

boundaries are predicted using the syllable information and word segmentation is preceded by syllable segmentation first (Haruechaiyasak et al, 2008). In Chinese, the characters (*hanzi*) are easily identifiable and segmentation can be done by tagging (Xue, 2003), or by using deep learning techniques to determine word-internal positions (Ma et al, 2018). In languages like Vietnamese, although the space delimiter is present, it can act either as a syllable separator or as a word separator (Nguyen et al, 2020).

2.2 In Sanskrit

Sanskrit Word Segmentation has been addressed from various aspects, from rule-based approaches to statistical and machine learning approaches. Hyman (2008) inferred that Finite State Automata can be used to model the *sandhi sūtras* (rules from *Pāṇini's Aṣṭādhyāyī*). Huet (2003) developed the SH, a tool that uses Finite State Automata to handle the non-determinism of resolving *sandhi* resulting into morphologically and lexically valid segments. This was further enhanced by introducing a graphical interface which presents all the possible lexically and morphologically valid segments (Goyal and Huet, 2016).

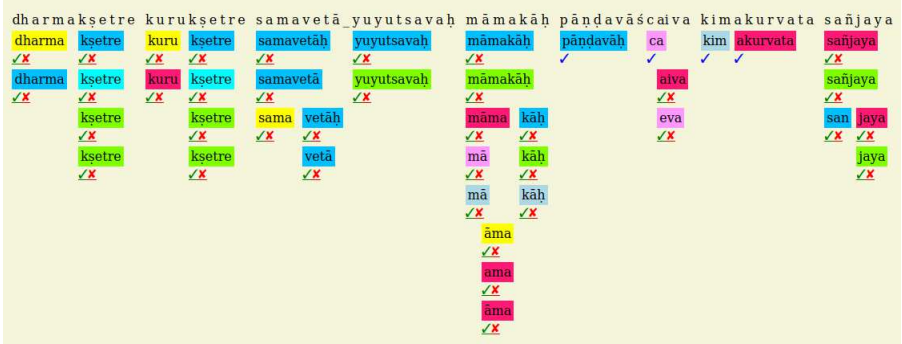
Mittal (2010) used OpenFST and augmented it with *sandhi* rules and finally validated the segments using optimality theory. Kumar et al (2010) developed a segmenter exclusively for Sanskrit compounds using probabilistic methods and optimality theory. Natarajan and Charniak (2011) proposed a statistical *sandhi* splitter using a Bayesian approach handling *sandhi* formations. Krishnan and Kulkarni (2019) proposed a ranking algorithm on SH (Huet, 2003) which scores each segmentation solution based on a prioritization with joint probabilities from word frequencies. Krishnan et al (2023) developed this ranking algorithm and incorporated it on top of SH's graphical interface (Goyal and Huet, 2016).

Hellwig (2016) proposed a deep learning model to detect the sentence boundaries. Krishna et al (2016) used the segments produced by the Heritage Segmenter and approached the selection of segments as a query expansion problem using Path Constrained Random Walks. Reddy et al (2018) proposed a sequence to sequence model without considering the linguistic features like lexical and morphological analysis. Aralikatte et al (2018) proposed another sequence to sequence model with two tasks: finding the split locations and then the individual splits. Hellwig and Nehrdich (2018) developed a character-level recurrent and concurrent neural network that works well with just the parallel corpus of unsegmented-segmented sentences. Krishna et al (2018) worked on a joint task of word segmentation and morphological tagging using an energy-based model. Krishna et al (2020) introduced a graph-based framework for structured prediction tasks in Sanskrit. Dave et al (2021) proposed an RNN based two-stage deep learning method for segmentation of isolated compound words without using any lexical or sentence information. Sandhan et al (2022) proposed a Transformer-based Linguistically Informed Sanskrit Tokenizer (TransLIST) for the task of segmentation and is the current state-of-the-art system.

3 Sanskrit Heritage Segmenter

The Sanskrit Heritage Engine is a platform that hosts a lexicon (The Sanskrit Heritage Dictionary) and various tools like segmenter, lemmatizer, declension and conjugation engines. The segmenter (referred to as SH in the paper) analyses any given text and segments it into all possible splits and displays them in a graphical interface where the user has the option to choose the required split. An example is shown in figure 2.

Fig. 2 Sanskrit Heritage Segmenter's Graphical Interface



It closely follows *Pāṇini's* system i.e., all the rules governing the concept of *sandhi* that occur in *Aṣṭādhyāyī* are taken into consideration. This segmentation is lexicon directed, using forms systematically generated from its own lexicon. It combines a fast segmentation algorithm using finite-state transducers and dynamic programming with a first-pass of chunking that limits the inherently exponential complexity to small-length chunks, making the whole segmentation analysis fast enough in practice to be usable interactively (Krishnan et al, 2023). SH has two modes for segmentation - Reader and Summary.

Reader mode: The Reader mode computes all possible segmentation solutions for a given text, and ranks them by assigning penalties based on shallow parsing. For each segmentation solution, it produces the segments, their possible morphological analyses and the corresponding *sandhi* that occurs with the subsequent segment.

Summary mode: The segmentation combined with the ranking in the Reader mode increases the time complexity and hence, the segmentation results are fed to a graphical interface (called Summary mode), where all the possible segments (extracted from all possible segmentation solutions) are populated on this interface along with options to choose or reject the segments. Upon selection or rejection of a segment, the interface updates itself dynamically by respectively choosing or pruning out the possibilities associated with the segment. Each segment is represented with the following features:

- word

- stem
- morphological analysis
- base
- derivational morphological analysis
- part of speech: indicated by colours of the segments, each colour corresponding to a part of speech (called as *phase* in SH). More details about phases are recorded in section 7.2.
- link to the lexicon entry for each stem and base

This fine-grained analysis helps the user to disambiguate and correctly pick the intended split and prune the solutions that are not required. Such information also helps in the further stages of sentential analysis like parsing, disambiguation, and discourse analysis.

Limitations: There are two major limitations of SH. One, the system does not consider *yogyatā* (meaning compatibility) in deciding the correct segmentation. Two, it does not automatically handle Out Of Vocabulary words. Although, for the unrecognized words, the interface allows the user to suggest the lemma (provided it is available in the Monier Williams Dictionary), which is then stored in the local dictionary for future use. The Reader mode does handle the over-generation problem to a certain extent by ranking the solutions based on a shallow parsing algorithm (Huet, 2007), but this is a primitive ranking mechanism and can be applied only on short sentences and sentences devoid of large compound words.

4 Datasets available for Word Segmentation

The previous decade saw various datasets released for the Word Segmentation task in Sanskrit. The widely used datasets are elaborated here.

4.1 SHMT

Sanskrit Hindi Machine Translation (SHMT) consortium⁸ developed a corpus for segmentation⁹. It has word-frequencies, compound-component frequencies and frequencies of *sandhi* between words and *sandhi* between compound components. Additionally, it also has a parallel dataset of compounds with their constituency analysis and compound types.

4.2 DCS - raw

The Digital Corpus of Sanskrit (DCS) (Hellwig, 2010–2021) is an annotated dataset, rich in morphological and lexical analysis, collected from around 250 Sanskrit texts. There are about 650,000 sentences with more than 4,500,000 word references and 175,000 unique words.

⁸A consortium under the funding from DeiT_Y (Department of Electronics and Information Technology), Govt. of India (2008-12)

⁹https://sanskrit.uohyd.ac.in/sci/GOLD_DATA/tagged-data.html

For every word reference, the following list of attributes is present in the DCS:¹⁰

1. lemma
2. morphological class
3. CNG value¹¹
4. preverbs (optional)
5. finite verb form (optional)
6. infinite verb form (optional)
7. position in sentence, position inside chunk
8. meaning of the lemma

Thus, its usage lies in various tasks in word segmentation, morphological parsing, word sense disambiguation, etc.

4.3 DCS - segmentation

The segmentation data used and released by Hellwig and Nehrdich (2018) contain the sentences from DCS along with the split points and segmented forms. As an example, table 2 presents the analysis of the sentence *pañca ratnāni mukhyāni coparatnacatuṣṭayam*.

Table 2 DCS Segmentation Analysis

Segmented form	Stem	Grammar	Sandhi Marking
<i>pañca</i>	<i>pañcan</i>	NUM	No
<i>ratnāni</i>	<i>ratna</i>	NC	No
<i>mukhyāni</i>	<i>mukhya</i>	JJ	No
<i>ca</i>	<i>ca</i>	CCD	Yes
<i>uparatna</i>	<i>uparatna</i>	NC	Yes
<i>catuṣṭayam</i>	<i>catuṣṭa</i>	NC	-

Sandhi occurs at two locations:

- (a) between *ca* and *uparatna* to form *coparatna* (*sandhi* across two words),
- (b) between *uparatna* and *catuṣṭayam* to form *uparatnacatuṣṭayam* (*sandhi* between the components of a compound word)

DCS marks both these *sandhis* but does not distinguish them. Looking at the Grammar tags for each of the segments, *ratnāni*, *uparatna* and *catuṣṭayam* all have been tagged as “NC” (indicating noun category). While *ratnāni*, a standalone word not involved in any *sandhi*, is to be marked as “NC”, *uparatna* should have been marked as “*iic.*” or “CPD” indicating an initial compound component. Although the base rules governing *sandhi*, either between words or between the compound components, are the same, there is a necessity to

¹⁰These values are present in the old DCS dump in the SQL format. DCS has shifted to a newer representation of this dataset in the CoNLL-U format. The present work takes both of these into consideration.

¹¹A value denoting the case, number, and gender of the given word, for nouns. Or the tense, aspect, person, number, and gaṇa (present system class) for verbs.

distinguish these two as the *sandhi* between the compound components predominantly occurs across the non-inflected forms while the *sandhi* across words occur over the inflected forms. Capturing the differences in the behaviour of inflected forms versus the non-inflected forms gives additional insights for segmentation, and also becomes crucial in the subsequent tasks such as parsing. However, this dataset is exclusively for word segmentation and hence compound splits are looked at in similar terms with word splits. Thus, differentiating the *sandhi* between words and the *sandhi* between compound components at the stage of word segmentation is debatable. And this gives us a single dataset for word segmentation, unlike the SHMT dataset which has a dedicated frequency list for compound components.

DCS is curated single-handedly and thus we can assume consistency in tagging. But the quality of this data tested on a small sample (Hellwig and Nehrdich, 2018) revealed that around 5.5% of the compound splits are doubtful and around 2% errors are due to segmentation.

4.4 SIGHUM - A Dataset for Word Segmentation

Krishna et al (2017) aligned the analyses of DCS' sentences with the possible analyses of SH to create a unified dataset. DCS' analyses for each of the sentences were represented as objects containing the sentence details like chunks, lemmas, and CNG values. A glimpse of what an object looks like is depicted in table 3.

Table 3 An example DCS Object data

Sentence Id	→	1
Sentence	→	<i>pañcaratnāni mukhyāni coparatnacatuṣṭayam</i>
Chunks	→	["pañcan", "ratna", "mukhya", "ca", "uparatna", "catuṣṭaya"]
Lemmas	→	[["pañcan"], ["ratna"], ["mukhya"], ["ca", "uparatna", "catuṣṭaya"]]
Morphological Class (CNG)	→	[["41"], ["41"], ["41"], ["2", "3", "31"]]

To match the two systems, data from SH's analysis was scrapped and certain parameters such as word, lemma, position, morphological information, chunk number, word length, and pre-verbs were extracted. Corresponding to the morphological analysis the CNG value was generated for the ease of alignment.¹² With all these parameters as attributes of the nodes, graphs were built for all the sentences. Standard graph processing libraries were used to extract the data from these graphs.

The XML based GraphML format was used to represent the candidate space segments. The GraphML files consist of graph structures, $G(V, E)$ as the representation for the analyses of a sentence. The nodes, V , are the possible

¹²CNG value is calculated from [cng-calculation](#)

splits, and the values in the edges, E , denote whether the participating nodes can co-exist in a solution or not i.e., whether or not they have an overlap in the position relative to the sentence, and that the overlapped portion does not follow any *sandhi* rule.

The two main parameters considered for aligning DCS and SH were *lemma* and *CNG value*. For every segment of DCS, a comparison was done with each of the nodes from SH, until a perfect match is obtained. There were multiple issues while aligning the segments, discussed by Krishna et al (2017) along with their respective solutions. Inconsistencies while dealing with *anusvāras* (nasal consonant) which should have actually been *anunāsikas* (homonasal consonants), were normalized. SH's stems are disjoint with the preverbs while DCS' stems are sandhied with the preverbs. Additional nodes were added with preverbed stems to handle this mismatch. Also, DCS' analyses is predominantly context-dependent and hence it prefers non-compositional analysis of compound wherever necessary. SH produces the non-compositional analysis only when it is available in its lexicon. In other cases, additional nodes were generated by sandhi-ing the components to match with the non-compositional analysis of DCS.

Thus around 119,000 sentences from the original DCS corpus were released in the SIGHUM dataset, which had the input sequence, ground truth segmentation, and morphological and lexical information about all the phonetically possible segments.

5 Differences between DCS and SH representation

Aligning the analyses of DCS and SH requires a detailed understanding on the similarities and differences of representations of the two systems. This section focuses on the issues in the representations at various levels such as chunk, word, stem, morphological analysis and compound, which hinder the process of alignment.

5.1 Differences at Chunk level

When two consecutive words undergo *sandhi* operation, then the following are the possible changes:

1. only the last character (and occasionally the last two characters¹³) of the first word undergoes change
2. only the first character of the second word undergoes change
3. both the last character(s) of the first word and the first character of the second word undergo changes

In all the three cases, SH considers the two words to be in a single chunk, whereas DCS treats them as a single chunk only in the third case, and in the

¹³In the case of *visarga-sandhi*, the last two characters undergo transformation or elision depending on the first character of the next word, particularly if we are using Non-Devanagari encoding scheme

first two cases, the two words constitute two different chunks. This results into the mismatch between the number of chunks. For example,

Unsegmented: *yan nābhishthitam*

Segmented: *yat na abhishthitam*

According to DCS, there are two chunks viz. “*yat*” and “*na abhishthitam*”, whereas SH treats them as a single chunk since *t* in *yat* is changed to *n* in the presence of the following *n*. This mismatch between the number of chunks affects chunk-by-chunk mapping of the two systems.

5.2 Segmented words

Another drawback of DCS’ representation is that the segmented words are not present separately¹⁴ and one has to construct the words from the available parameters vis-à-vis lemma and inflectional morphology. But, there could be multiple possible word-forms generated from the same lemma and morphological analysis. For example, the root *gup* (to guard) is derived with gerundive suffixes (*yat*, *anīyar*, *tavya*) to form respectively, *gopya*, *gopanīya* and *gopitavya* (all the three indicating the sense of “to be guarded / preserved”, the difference being the usage based on context). DCS marks all the three forms as “Ger” (gerundives). With only the root *gup* and the morphological tag “Ger”, it is not trivial to single out one of the three word-forms. Such information missing at the levels of stem and morphological analysis become crucial for word generation.

5.3 Stem level differences

1. In the case of derived words¹⁵, DCS is not uniform in marking the analyses. Sometimes, it marks both the inflectional as well as the derivational analyses and sometimes only one of them. SH, on the other hand, marks both the analyses uniformly. This may result into one to many mapping if the words are ambiguous at inflectional / derivational level. For example, let us consider the word *hitam*. DCS stores only the inflectional analysis (iic.) with the stem *hita*, while SH provides the base also (*hi* or *dhā*). Interestingly, there are four dictionary entries in DCS for the form *hita*, but none of them have any information regarding the base form. Thus, automatic alignment is not possible with just the available DCS parameters. The design decision to mark either the derived lemma or the base lemma depending on the context results into a non-uniform annotation and it also suppresses derivational information which could play an important role in the tasks of compound analysis or sentential analysis. Such cases are recorded separately for discussion.

¹⁴In the latest updates of DCS, the segmented forms have been introduced. But these are generated using the neural network Word Segmentation model proposed by [Hellwig and Nehrdich \(2018\)](#) and hence there could be errors in the segmentations.

¹⁵In Sanskrit, a root (*dhātu*) is derived into a stem (*prātipadika*) by adding a few affixes.

2. **Handling homonymous stems:** This involves mapping the meaning spaces of DCS and SH. For example, the stem *siddha* has two senses in SH's lexicon with two completely different meanings. One indicating accomplishment, realization, perfection and the other indicating prevention or pushing aside. On the other hand, DCS has four entries with varied meanings. It is not trivial to map the two senses of SH with the four senses of DCS and hence during the alignment process, such multiple senses were collapsed together.
3. **Handling the iics of compounds:** In the case of the initial component of a compound, SH marks the stem whereas DCS marks the surface form itself as the stem. For example, *mahā* in the compound *mahāyoga*, is analysed by SH with stem *mahat* while DCS marks the iic form *mahā* as the stem.
4. **Handling the derived verbal stems:** In the case of derived verbal stems such as causative forms, DCS provides the analysis with the causative verbal form as the stem while SH provides the base verb stem with “causative” as a feature in the analysis. For example, for the word *gamayati* (sends), DCS marks the derived verbal form *gamay* (to send) as the stem while SH marks the underived verbal form *gam* (to go) as the stem and “causative” is marked as a feature along with other features viz. tense (present active), person (third), number (singular).

5.4 Differences at the level of Morphological Analysis

As far as the inflectional analysis of nouns is concerned, barring the nomenclatures, the analysis of DCS and SH match perfectly. Regarding the verbal inflectional analysis, we have the following observations:

- SH provides the information of the set of suffixes (*parasmaipada* (active voice) vs *ātmanepada* (reflective / middle voice)) and the class to which the verb belongs to. This information is crucial from the disambiguation point of view since the same verbal root with different suffixes or different classes would have different meanings.
- In the case of primary derivatives,¹⁶ SH provides more information such as the specific suffixes added. For example, the conjugational suffix in the case of derived verbs such as causative and desiderative forms. DCS on the other hand does not mark this information resulting into one-many mapping from DCS to SH.

5.5 Compounds

In addition to the issue with iics of compounds (section 5.3), DCS and SH also differ in respect to the compositional analysis of a compound. Compounds which are named entities are not decomposed by DCS. Since SH analyses words without referring to the context, it produces analyses by decomposing the compounds into its components. This results into mismatches.

¹⁶A primary derivative is a nominal obtained from a verbal root and is termed as *kṛdanta*.

All the differences discussed above are due to the design decisions of DCS and SH and are to be considered before aligning the two systems. What follows is the alignment process carried out keeping in mind these differences.

6 Alignment of DCS-SH

Three alignment approaches were carried out.

- (a) The first alignment process is in essence the same as that described in SIGHUM (section 4.4). The only difference is that an improved version of SH is considered and the differences in the representation of linguistic information was given more importance. This resulted into the Alignment 1 dataset. The alignment process and the linguistic observations are reported in Krishnan et al (2023). This dataset was released partially for the Word Segmentation and Morphological Parsing for Sanskrit - Hackathon.¹⁷ This will be referred to as the Hackathon dataset in the paper.
- (b) In 2022, DCS released the improved version of the data in CoNLL-U format.¹⁸ SH was also improved further taking into consideration the differences reported in Krishnan et al (2023). Hence the alignment process was carried out **with and without** considering the **differences due to chunk-unit** in both SH and DCS, respectively resulting into the **second and third** alignments.

For all the three approaches, alignment was done in four steps as follows:

0. For every sentence of DCS, extract DCS data and SH analyses,
 1. Convert the extracted data into a normalized form,
 2. Compare DCS entry with all possible analyses of SH to get the aligned analysis,
 3. From the entries where the alignment was not possible, handle the mismatches due to compounds ignoring compositionality and rerun the alignment process

6.1 Step 0: Extraction of DCS and SH Analyses

6.1.1 Extracting DCS data

There are two major versions of DCS. The first is the old DCS dump in the SQL format.¹⁹ The second version is the more recent dataset released in the CoNLL-U format.²⁰

SIGHUM represented DCS SQL analyses as Pickle objects (table 3). These objects were converted to a JSON format for the first alignment (table 4). For the second and third alignments, the CoNLL-U representation of DCS was converted to a JSON representation (table 5) for ease of analysis.

¹⁷<https://sanskritpanini.github.io/>

¹⁸<https://universaldependencies.org/docs/format.html>

¹⁹<https://drive.google.com/open?id=1zKHtrnRTqW6TroOoepFgTGBsPT9D6i6k>

²⁰<https://github.com/OliverHellwig/sanskrit/tree/master/dcs/data/conllu>

Table 4 DCS data in JSON format for First Alignment

Sentence Id	→	1
Sentence	→	<i>ādiśvarāya praṇamāmi tasmai yenopadiṣṭā haṭhayogavidyā</i>
Lemmas	→	[[“ <i>ādi</i> ”, “ <i>īśvara</i> ”], [“ <i>praṇam</i> ”], [“ <i>tad</i> ”], [“ <i>yad</i> ”, “ <i>upadiṣṭa</i> ”], [“ <i>haṭhayoga</i> ”, “ <i>vidyā</i> ”]]
Morph Code	→	[[“3”, “109”], [“-11”], [“109”], [“89”, “30”], [“3”, “30”]]
Morphs	→	[[“m”, “m”], [“1. Ā.”], [“pron”], [“pron”, “6. Ā.”], [“m”, “f”]]
Der Lemmas	→	[[“”, “”], [“”, “”], [“”, “ <i>upadiṣ</i> ”], [“”, “”]]
Der Morph Code	→	[[“”, “”], [“”, “”], [“”, “-190”], [“”, “”]]
Prefixes	→	[[“”, “”], [“ <i>pra</i> ”], [“”, “”], [“”, “ <i>upa</i> ”], [“”, “”]]

6.1.2 Extracting SH analyses

Figure 2 showed the web-based representation of the SH analyses in the form of a graphical interface. SIGHUM scrapped through the web page results to extract the analyses. The first alignment followed a similar approach and extracted the parameters such as word, lemma, position, morphological information, chunk number, word length, and pre-verbs. This was represented as a graph in the GraphML format with the segments as nodes, segment parameters as the node attributes, and the edge values indicating the co-occurrence of the participating nodes in a solution.

For the second and third alignments, SH was internally updated to produce the list of all segments and their analyses in a .tsv format where each of the fields have the values according to table 6.

*phase*²¹ and *phase_color*²² have been included for the subsequent tasks of compound analysis. Mapping SH’s sense with DCS’ sense is not trivial and hence the sense fields have a comma-separated list of senses which is kept for use in the subsequent tasks like homonymy disambiguation. Hence, *phase*, *phase_color*, *derived_stem_sense* and *base_sense* are not used in the alignment process.

chunk_id, *pos_in_chunk*, *derived_stem*, *inflectional_morph_analysis*, *base* and *derivational_morph_analysis* are the primary parameters for comparison in the alignment process, where *chunk_id* and *pos_in_chunk* are considered for alignments 1 and 2 only.

²¹Phase roughly corresponds to the Part Of Speech with grammatical information.

²²Each phase is assigned a color by SH for clarity in visual representation. Sometimes, multiple phases are assigned the same color when their functionalities are similar. For example, nominal stems inflected with case suffixes (*vibhaktis* - nominative, accusative, etc. except vocatives) are assigned the Phase “Noun” and the nominal stems derived from roots by adding primary derivative suffixes (*kṛdantas*) are assigned the Phase “Krid”. But both have been assigned the color “Deep_sky” indicating that at the level of Phases (parts of speech), their functionality is similar and hence *phase_color* is separately recorded.

Table 5 DCS data in JSON format for Second and Third Alignments

text	→	<i>gheraṇḍasaṃhitā</i>
text ID	→	1
chapter	→	ghers, 1
chapter ID	→	270
sent ID	→	1
joint sentence	→	<i>ādīśvarāya praṇamāmi tasmai yenopadiṣṣā haṭhayogavidyā</i>
unsegmented form	→	[<i>“ādīśvarāya”, “praṇamāmi”, “tasmai”, “yenopadiṣṣā”, “haṭhayogavidyā”</i>]
stem	→	[<i>“ādi”, “īśvara”</i>], [<i>“praṇam”</i>], [<i>“tad”</i>], [<i>“yad”, “upadiṣṭa”</i>], [<i>“haṭhayoga”, “vidyā”</i>]
stem ID	→	[<i>“57973”, “64099”</i>], [<i>“162415”</i>], [<i>“37875”</i>], [<i>“37877”, “160400”</i>], [<i>“121886”, “121245”</i>]
morph	→	[<i>“Case=Cpd”, “Case=Dat Number=1 Gender=Masc”</i>], [<i>“Tense=Pres Mood=Ind Person=1 Number=Sing”</i>], [<i>“Case=Dat Number=1 Gender=Masc”</i>], [<i>“Case=Ins Number=1 Gender=Masc”</i>], [<i>“Case=Nom Number=1 Gender=Fem VerbForm=PPP”</i>], [<i>“Case=Cpd”, “Case=Nom Number=1 Gender=Fem”</i>]]
grammar	→	[<i>“m”, “m”</i>], [<i>“1. Ā.,1.P.”</i>], [<i>“pron”</i>], [<i>“pron”, “6.P.,6. Ā.”</i>], [<i>“m”, “f”</i>]]
xpos	→	[<i>“NC”, “NC”</i>], [<i>“V”</i>], [<i>“PRD”</i>], [<i>“PRL”, “PPP”</i>], [<i>“NC”, “NC”</i>]]
preverbs	→	[<i>“”, “”</i>], [<i>“pra”</i>], [<i>“”</i>], [<i>“”, “upa”</i>], [<i>“”, “”</i>]]

6.2 Step 1: Normalization of the Analyses

The comparison on the design decisions of the two systems (section 5) showed that there were many differences and each has to be considered prior to the process of alignment to make sure that the representations of both the systems are normalized.

6.2.1 Aligning Stems

- **Base vs Derived stem:** SH provides both the base and the derived stems while DCS sticks to one stem that is more apt in the given context. For the word *gopaniyā*, DCS marks the stem as *gup*, while SH produces additionally the derived stem *gopaniyā*. In such cases, the DCS stem is aligned with either the derived stem or the base stem of SH, whichever is available in DCS, but both the derived and the base stems of SH are considered for the final dataset.

Table 6 Reference for the fields in SH's analysis

field_id	value	field_id	value
1	segment_id	8	derived_stem_sense
2	word_form	9	inflectional_morph_analysis
3	chunk_id	10	base
4	pos_in_chunk	11	base_sense
5	phase	12	derivational_morph_analysis
6	phase_color	13	pre_verb
7	derived_stem		

- **Mapping pronouns:** SH uses the *Pāṇinian* system for pronouns' stems while DCS uses a form that is common across all the surface forms. For example *yuṣmad* (SH) - *tvad* (DCS). These are normalized to the SH format.
- **Mapping derived verbal stems:** causative, desiderative and intensive forms are stored with modified stems in DCS without providing its derivation while SH stores the base stem with a morphological feature such as causative (ca.), desiderative (des.) or intensive (int.). For example, *gopayet* (hide something) is analysed by DCS with the modified stem *gopay*, while SH uses the root *gup* (to guard / hide) along with the morphological feature "ca." (causative).
- **Handling Nasal Consonant (*Anusvāra*):** In both DCS and SH, the forms with *anusvāra* (nasal consonant) are normalized to the corresponding *anunāsika* (homonasal consonant). Similarly in some verbal non-finite forms, reduplication of a consonant is observed optionally. In all such cases the forms are normalized to the ones without reduplications.
- **Handling iics:** SH and DCS produce different stems for the *iic* component of compounds. These are also normalized to the forms produced by SH for alignment purpose.
- **Handling Preverbs:** DCS produces stems sandhied with their preverbs. For the first alignment, the stems of SH were disjoint with the preverbs and hence preverbs were sandhied with the stems. For example, *pra-nam* → *praṇam* for the word *praṇamāmi*. For the second and third alignments, the .tsv format produces the stems sandhied with their corresponding preverbs.

6.2.2 Aligning Morphological Analyses

- **Morphological Analysis Mapper:** Since the representations of the morphological analysis is different in the two systems, a one-to-one mapping is required to compare the analyses of both the systems. For the first

alignment, a look-up table was constructed with the keys being unique CNG entries and the values being SH's morphological analysis.²³ For the second and third alignments, a morphological analysis table was built to map the morphological analysis of SH with that of DCS.²⁴

- **Inflectional vs Derivational Analysis:** It is observed that DCS provides the derivational analysis only when the context requires it. Hence, the first preference for comparison is given to the default inflectional analysis alone. If the alignment fails with the inflectional analysis, then the derivational analysis is taken into consideration along with the inflectional analysis, provided the stem marked by DCS is mapped with SH's base stem. For example, the word *hitam* has the stem *hita* and morphological analysis "Case=Nom|Number=1|Gender=Neut" but SH produces the derivational analysis as well ("pp."). In this case, the inflectional analysis alone is considered for the alignment. For the word *gopanīyam*, DCS marks both the inflectional as well as the derivational form as "Case=Nom|Number=1|Gender=Nuet|VerbForm=Ger". DCS' stem (*gup*) matches with the SH's base stem and hence both the inflectional and the derivational morphological analyses are considered together for the alignment.
- **Handling Missing morphological Analyses:** There are about 4.5 million word references in DCS and about 17% of these do not have the morphological analysis marked. It was observed that about 90% of these are indeclinables. A list of possible indeclinables was prepared prior to the alignment process. During the alignment, these are referred to for comparison with the stems which do not have the morphological analysis marked.

6.3 Step 2: Comparison for Alignment

For every sentence, DCS' ground truth analysis (stem and morphological analysis) of a segment is compared with the analysis of every segment produced by SH. DCS' stem is compared with the base and derived stems of SH. DCS' morphological analysis is compared with the inflectional and derivational morphological analysis of SH, using the CNG look-up table (in Alignment 1) and the Morphological Analysis Mapper (in Alignments 2 and 3).

For the first and the second alignments, *chunk_id* and *position_in_chunk* help in an in-position comparison where the stem and morphological analysis from DCS are aligned with the SH analyses at exactly the same position in the sentence as proposed by DCS. It was observed in section 5 that DCS and SH differ in their definitions of a chunk and how it affects the chunk-based in-position alignment process. To circumvent this drawback, this restriction of in-position comparison was lifted off in the third alignment.

Table 7 shows the results after Step 2 of the alignment process in the three approaches:

²³The look-up table for DCS' CNG and SH analyses is available at: [dcs-cng-sh-mapping](#)

²⁴The morphological analysis mapper is available at: [dcs-sh-morph-mapping](#)

Table 7 Alignment Results Step 2

	A1 ¹	A2 ²	A3 ³
Number of sentences	621,445	621,327	621,327
Sentences Unrecognized by SH	* ⁴	145,313	145,313
Sentences Recognized by SH	*	476,014	476,014
Aligned sentences	73,429	112,738	160,453
Multiple alignments	18,100	7,313	10,996
Missed Alignments	503,761	355,963	304,565

¹Alignment 1, used the DCS SQL Dump and reported in [Krishnan et al \(2023\)](#)

²Alignment 2, used the DCS CoNLL-U data

³Alignment 3, used the DCS CoNLL-U data (ignoring chunk boundaries)

⁴* - data unavailable

The following observations are recorded after Step 2:

- The alignment is much stricter when the chunk-level comparison is present and hence the number of sentences with complete alignment is less in Alignments 1 (73,429) and 2 (112,738) when compared to 3 (160,453).
- With the recent updates to SH and the modification of the alignment algorithm, the number of aligned sentences increased from Alignment 1 to Alignments 2 and 3.
- Sentences falling under Multiple Alignments have one to many correct mappings of DCS to SH analyses and hence it is not trivial to disambiguate them.
- The sentences under Missed Alignments are then considered for further processing in Step 3.

6.4 Step 3: Alignment on the Updated analyses

The DCS and SH analyses of the sentences under Missed Alignments are modified as follows:

1. If a missed out segment has causative forms, then the causatives to roots mapping is updated
2. The iics and corresponding ifcs of compounds in the SH analyses are combined to form possible compounds with non-compositional analysis. For example, *haṭha* and *yoga* of SH analysis are combined to form a new segment with the new stem being *haṭhayoga* and the chunk_ids taken from the first segment(*haṭha*) and the morphological analyses taken from that of the second segment (*yoga*).

Finally, the process of re-alignment is carried out once again on the mis-aligned entries with the modified analyses.

6.5 Alignment Results

Table 8 shows the comparison of observations in Step 2 versus Step 3. And table 9 shows the summary of the alignment process in all the three approaches.

Table 8 Alignment Comparison Steps 2 and 3

	A1¹		A2²		A³	
	Step 2	Step 3	Step 2	Step 3	Step 2	Step 3
All Sentences	621,445	338,305	621,327	163,221	621,327	151,137
Unrecognized	* ⁴	*	145,313	0	145,313	0
Recognized	*	*	476,014	0	476,014	0
Aligned	73,429	56,921	112,738	34,801	160,453	30,718
Multiple	18,100	66,369	7,313	26,865	10,996	18,312
Missed	529,916	215,015	355,963	101,555	304,565	102,107
Modified	338,305	*	163,221	*	151,137	*

¹A1 - Alignment 1

²A2 - Alignment 2

³A3 - Alignment 3

⁴* - data unavailable

Table 9 Overall Alignment Observations

	A1¹	A2²	A3³
All Sentences	621,445	621,327	621,327
Aligned	130,350	147,539	191,171
Multiple	84,469	34,178	29,308
Missed⁴	406,626	439,610	400,848

¹A1 - Alignment 1

²A2 - Alignment 2

³A3 - Alignment 3

⁴Includes those sentences which were not recognized by SH, those sentences which were not aligned in step 3 and neither modified in step 3

The following are the achievements of the alignment processes:

1. Parallel Corpus of unsegmented-segmented sentences

2. List of all aligned (word, lemma, cng, morph) tuples and their frequencies from Alignment 1 and list of all aligned (word, stem, base, derived_morph, base_morph) tuples and their frequencies from Alignments 2 and 3.
3. List of all *sandhi* rules and their frequencies
4. In the first alignment, the edge values in the GraphML files had been updated further to have value 3 between the segments which are a part of the ground truth segmentation. In addition to this, unified datasets with attributes from both the DCS and SH analyses (in JSON and GraphML format from Alignment 1 and JSON format from Alignments 2 and 3) were built. A sample of the unified dataset from Alignment 1 is shown in table 10. The JSON format of the same sentence after alignments 2 and 3 are shown in table 11. In addition to the keys shown here, additional keys like upos, xpos, semantic_ids, dependency relations, which are useful for subsequent tasks are present along with those which correspond to the metadata of the sentence like text, text_id, chapter_id, sent_counter.
5. We can observe a significant increase in the number of aligned sentences from Alignments 1 to 3. But the number of sentences with multiple analyses has reduced. This shows that the addition of linguistic parameters and the removal of chunk boundaries have helped towards the alignment of sentences with multiple analyses.

6.6 Alignment Mismatches

We observe from table 8 that in all the three alignment approaches, there was a significant number of sentences that went unaligned in both step 2 and 3. The common causes of mis-alignment are recorded here.

Non-recognition by SH: The first case deals with the Out Of Vocabulary words (unrecognized words) and amounts to about 25% of the DCS sentences. Secondary derivatives (*taddhita* forms) are recognized by SH only when their stem is lexicalised. And there could be a huge list of domain-specific named entities among these unrecognized words. The required solution would be to update SH's lexicon with the missing entries.

Incorrect segmentation by SH: Some forms are not analysed by SH resulting into incorrect segmentations. This could be due to issues in resolving *sandhi*, mostly for sentences from Vedic literature where preverbs are disjoint with their corresponding verbal forms. Incorrect segmentations could also be caused from non-recognition and missed out morphological analyses of certain forms. For example, the sentence *tarjanīmādhyaṃānāmā aṅgulitrayayogataḥ* has the analysis (stems) from DCS as *tarjanī-madhyamā-anāman aṅguli-traya-yoga* but SH could analyse only the stems *tarjanī* and *aṅguli*. *madhyamā* is not analysed as an *iic.* and *aṅguli-traya-yogataḥ* is not recognized as a compound. Compositionality (discussed ahead) also plays a role in deciding the correct segmentation in this example.

One-to-many mapping: The mapper between the morphological analyses of DCS and that of SH has a one to many mapping as DCS suppresses certain information while SH explicitly specifies such information. For example,

Table 10 Alignment 1 Results format in JSON

key	value
sent_id	14,924
sentence	<i>vāyuh prakupito yasya rūkṣāhārasya dehinaḥ</i>
word	[[“vāyuh”], [“prakupitaḥ”], [“yasya”], [“rūkṣa”, “ahārasya”], [“dehinaḥ”]]
stem	[[“vāyu”], [“prakupita”], [“yad”], [“rūkṣa”, “ahāra”], [“dehin”]]
sense	[[“1”], [“1”], [“1”], [“1”, “1”], [“1”]]
morph	[[“m. sg. nom.”], [“m. sg. nom.”], [“m. sg. g.”], [“iic.”, “m. sg. g.”], [“m. sg. g.”]]
cng	[[“29”], [“29”], [“149”], [“3”, “149”], [“149”]]
base_stem	[[“”], [“prakup”], [“”], [“”, “”], [“”]]
base_sense	[[“0”], [“1”], [“0”], [“0”, “0”], [“0”]]
base_morph	[[“”], [“pp.”], [“”], [“”, “”], [“”]]
base_cng	[[“”], [“-190”], [“”], [“”, “”], [“”]]
pre_verb	[[“”], [“pra”], [“”], [“”, “”], [“”]]
graphml_node_ids	[[“47”], [“54”], [“34”], [“52”, “36”], [“39”]]

the secondary conjugations viz. causative(*ṇic*), desiderative(*san*) and intensive(*yañ*) are explicitly specified with keys “ca.”, “des.”, and “int.” by SH but these are suppressed by DCS. The information of class (*gaṇa*) is important to decide the meaning of the verb. The information of active and passive voice is needed for deciding the dependency structure of a sentence. These are not specified in DCS analyses.

Compositionality: This is a more challenging case where DCS has an explicit entry for the non-compositional analysis of certain compounds, whereas SH sticks to the compositional analyses²⁵. Compositionality depends on the individual meanings of the combining entities and also the context. The word *pītāmbaram* when referred to as the deity *Viṣṇu*, takes the non-compositional meaning, and when referred to as the yellow cloth, it has a compositional meaning (from its constituents *pīta* (yellow) and *ambaram* (cloth)). DCS provides only the non-compositional analyses of certain compounds where the context requires the non-compositional meaning. On the other hand, SH produces the non-compositional analyses if the lexicon has

²⁵with a few exceptions where certain named entities have been lexicalised with non-compositional analysis

Table 11 Alignment 2 and 3 Results format in JSON

key	value
sent_id	"14924"
joint_sentence	" <i>vāyuh prakupito yasya rūkṣāhārasya dehinaḥ</i> "
position	[[<i>"1"</i>], [<i>"2"</i>], [<i>"3"</i>], [<i>"4"</i>], [<i>"5"</i>], [<i>"6"</i>]]
unsegmented_form	[<i>"vāyuh"</i> , <i>"prakupito"</i> , <i>"yasya"</i> , <i>"rūkṣāhārasya"</i> , <i>"dehinaḥ"</i>]
sh_word	[[<i>"vāyuh"</i>], [<i>"prakupitaḥ"</i>], [<i>"yasya"</i>], [<i>"rūkṣa"</i>], [<i>"āhārasya"</i>], [<i>"dehinaḥ"</i>]]
word	[[<i>"vāyuh"</i>], [<i>"prakupitaḥ"</i>], [<i>"yasya"</i>], [<i>"rūkṣa"</i>], [<i>"āhārasya"</i>], [<i>"dehinaḥ"</i>]]
phase	[[<i>"Nouc"</i>], [<i>"Kric"</i>], [<i>"Pron"</i>], [<i>"Iicc"</i>], [<i>"Nouv"</i>], [<i>"Nouc"</i>]]
phase_color	[[<i>"Deep_sky"</i>], [<i>"Deep_sky"</i>], [<i>"Light_blue"</i>], [<i>"Yellow"</i>], [<i>"Deep_sky"</i>], [<i>"Deep_sky"</i>]]
dcs_stem	[[<i>"vāyu"</i>], [<i>"prakup"</i>], [<i>"yad"</i>], [<i>"rūkṣa"</i>], [<i>"ahāra"</i>], [<i>"dehin"</i>]]
sh_stem	[[<i>"vāyu"</i>], [<i>"prakupita"</i>], [<i>"yad"</i>], [<i>"rūkṣa"</i>], [<i>"ahāra"</i>], [<i>"dehin"</i>]]
sh_stem_sense	[[<i>"1"</i>], [<i>"1"</i>], [<i>"1"</i>], [<i>"1"</i>], [<i>"1"</i>], [<i>"1"</i>]]
stem	[[<i>"vāyu"</i>], [<i>"prakupita"</i>], [<i>"yad"</i>], [<i>"rūkṣa"</i>], [<i>"ahāra"</i>], [<i>"dehin"</i>]]
sh_base	[[<i>"</i> "], [<i>"prakup"</i>], [<i>"</i> "], [<i>"</i> "], [<i>"</i> "], [<i>"</i>]]
sh_base_sense	[[<i>"</i> "], [<i>"1"</i>], [<i>"</i> "], [<i>"</i> "], [<i>"</i> "], [<i>"</i>]]
base	[[<i>"</i> "], [<i>"prakup"</i>], [<i>"</i> "], [<i>"</i> "], [<i>"</i> "], [<i>"</i>]]
dcs_morph	[[<i>"Case=Nom Number=1 Gender=Masc"</i>], [<i>"Case=Nom Number=1 Gender=Masc VerbForm=PPP"</i>], [<i>"Case=Gen Number=1 Gender=Masc"</i>], [<i>"Case=Cpd"</i>], [<i>"Case=Gen Number=1 Gender=Masc"</i>], [<i>"Case=Gen Number=1 Gender=Masc"</i>]]
sh_morph	[[<i>"m. sg. nom."</i>], [<i>"m. sg. nom."</i>], [<i>"m. sg. g."</i>], [<i>"iic."</i>], [<i>"m. sg. g."</i>], [<i>"m. sg. g."</i>]]
sh_base_morph	[[<i>"</i> "], [<i>"pp."</i>], [<i>"</i> "], [<i>"</i> "], [<i>"</i> "], [<i>"</i>]]
pre_verb	[[<i>"</i> "], [<i>"pra"</i>], [<i>"</i> "], [<i>"</i> "], [<i>"</i> "], [<i>"</i>]]
xpos	[[<i>"NC"</i>], [<i>"PPP"</i>], [<i>"PRL"</i>], [<i>"JJ"</i>], [<i>"NC"</i>], [<i>"NC"</i>]]

such an entry, while it produces the compositional analysis always. It is tedious to update SH's lexicon with all such non-compositional forms as it could also result into over-generation of the lexicon entries.

This was handled partially by generating all possible combinations of the grouping of compound components, which are then compared with the DCS entries. Some of the compounds with a huge number of components were ignored as this part of the algorithm was time consuming since the possible groupings of compound components is a Catalan number, C_n ,

$$C_n = \frac{(2n)!}{(n+1)! n!} \quad (1)$$

if there are n components in a compound (Huet, 2009).

7 SH Segmenter: Ranking Solutions

Availability of a well annotated linguistically rich dataset was a trigger for improving the performance of segmentation algorithms. Krishnan et al (2023) used the dataset from Alignment 1 for ranking the multiple solutions produced by SH. Li and Girrbach (2022) used the Hackathon dataset for the joint task of word segmentation and morphological parsing in the WSMP Hackathon. Li and Girrbach (2022) showed a comparison of the results with other models in the same hackathon on the individual tasks of word segmentation and morphological parsing, where the F1 scores for word segmentation reached as high as 97.478. Sandhan et al (2022) used the same Hackathon dataset and compared it with the SIGHUM dataset for the task of word tokenization using a Transformer-based Linguistically Informed model, and it showed that both the SIGHUM and Hackathon datasets produce very similar performance scores in terms of Precision, Recall and F1 score with the models performing slightly better when using the SIGHUM dataset.

SH produces all the possible segmentations with their corresponding morphological analyses. In order to rank these segmentations and pick the best one among all possible segmentations, the results of the alignment process, specifically the frequencies extracted from the parallel corpus of unsegmented-segmented sentences, were used.²⁶ We now describe the ranking algorithm below.

7.1 Augmenting SH Segmentation with ranking

SH has two segmentation algorithms - the original algorithm that produces the segmentation in a list mode (called Reader), and the improved algorithm that produces the output in a graph mode (called Summary). The Reader does the segmentation first and has an in-built ranking mechanism based on shallow dependency analysis. The Summary mode displays all the segments in

²⁶The code for extracting the frequencies from the alignment datasets and the resultant frequencies with different metrics such as word, transition and stem-morph are available here: [alignment_dataset](#)

the form of a graphical interface where users can choose or reject the segments. We now describe the modifications done to these two modes.

7.1.1 The Reader Mode

The results in the Reader mode are listed out with the segments, their part-of-speech and lexical analyses, and the subsequent *sandhi* (called transition) that happens at the word/component boundaries. [Krishnan and Kulkarni \(2019\)](#) proposed a modification to the Reader mode where the information about part-of-speech and lexical analyses is ignored, and the segmented words alone are considered for the solutions. A ranking algorithm was introduced which prioritizes the solutions based on a joint probability calculated from word and transition frequencies of the SHMT corpus. The joint probability was calculated with two metrics.

The first metrics considered the unigram probabilities of the segments (words) and the bigram probabilities of the transitions and a cumulative product across all the segments was calculated as:

$$C_{total} = \prod_{i=1}^n P_{w_i} \times P_{t_i} \quad (2)$$

where P_{w_i} is the word probability and P_{t_i} is the transition probability.

The second metrics considered the *word-probability* alone where the resultant value is the product of unigram probabilities of the segments:

$$C_{total} = \prod_{i=1}^n P_{w_i} \quad (3)$$

These metrics of joint probabilities of the segments and transitions are similar to First Order Markov Model except that the segments are not states and hence the probability of the segment does not depend on the previous segment. The solutions with higher joint probabilities combined with the least number of segments are ranked higher.

This modification to the Reader mode was found to be inefficient in the case of long sentences and sentences with long compounds. SH has already treated these issues in its Summary mode. Therefore the ranking algorithm was incorporated in SH's Summary mode.

7.1.2 The Summary Mode

Segmentation in SH has two stages:

- 1. Chunking:** The sentence is initially divided into chunks based on certain *sandhi* rules and the default spaces provided in the input sentence. This reduces the segmentation task into chunk-level segmentation and increases exponentially the speed of the segmentation process.

- 2. Segmentation:** Each chunk is analysed further where segmentation and morphology recognition are done parallelly, using finite state transducers

and dynamic programming and the morphologically valid segments are finally registered into the graphical interface.

This algorithm developed by Goyal and Huet (2016) was modified in Krishnan et al (2023) by adding dovetailing to rank the solutions and maintain a bucket of best n solutions²⁷ based on the probabilities obtained from Alignment 1. Thus, in addition to the two steps of the segmentation, two more steps had been added to rank the solutions.

1. Constructing the list of chunk segmentations
2. Dovetailing mechanism over the list of chunk segmentations to keep a list of best segmentations

This makes sure that the less probable segments are omitted and only the most probable segments are considered for the final result.

7.2 Criterion for Ranking

SH in addition to providing possible ways to segment a string, also provides the morphological analyses of the components involved. Both these features are useful from the disambiguation point of view. For example,

(a) The string *śvetodhāvati* is ambiguous between *śvā itaḥ dhāvati* (A dog runs from here) and *śvetaḥ dhāvati* (The white one runs). Though both of these segmentations are meaningful, the second one is more frequent in Sanskrit than the first. And our segmenter should rank these solutions accordingly.

(b) The string *rāmovanaṅgacchati* can be split in only one way viz. *rāmaḥ vanam gacchati*. However each of these three words have more than one morphological analyses (table 12). The SH segmenter instead of providing only one segmentation, provides four different solutions with combinations presented in table 13.

Table 12 An example of multiple morphological analyses

Segment	Analysis
<i>rāmaḥ</i>	noun → masculine singular nominative of stem <i>rāma</i> verb → present active first person plural of root <i>rā</i> (of class 2)
<i>vanam</i>	noun → neuter singular accusative of stem <i>vana</i> noun → neuter singular nominative of stem <i>vana</i>
<i>gacchati</i>	verb → present active third person singular of root <i>gam</i> (of class 1) present participle → masculine singular locative of root <i>gam</i> (of class 1) present participle → neuter singular locative of root <i>gam</i> (of class 1)

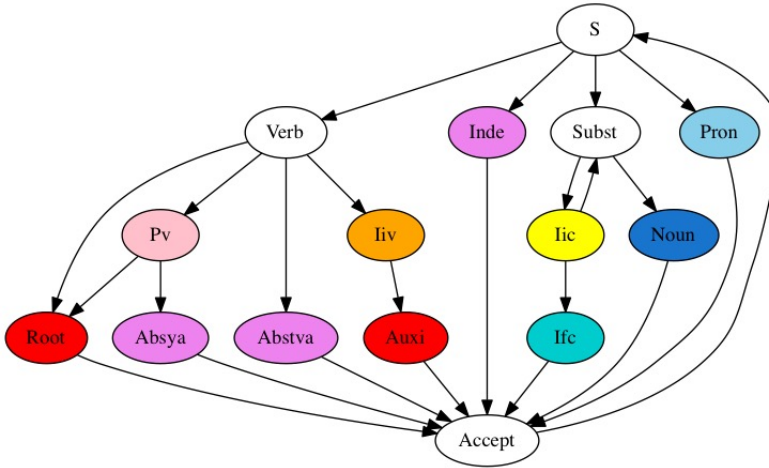
Each of these is further ambiguous since the word *vanam* and also *gacchati* as a present participle have multiple morphological analyses, leading to 12 possible unambiguous solutions. But SH segmenter clubs the morphological analyses in a single *phase* where it cannot be disambiguated structurally, and hence only four solutions. Phases are similar to part of speech tags

²⁷ n is internally fixed as 5, 10, 50 or 100

Table 13 SH segmentation solutions for *rāmovanaṅgacchati*

No.	Segment	Analysis
1	<i>rāmaḥ</i> <i>vanam</i> <i>gacchati</i>	noun → masculine singular nominative of stem <i>rāma</i> noun → neuter singular accusative / nominative of stem <i>vana</i> verb → present active third person singular of root <i>gam</i> (of class 1)
2	<i>rāmaḥ</i> <i>vanam</i> <i>gacchati</i>	noun → masculine singular nominative of stem <i>rāma</i> noun → neuter singular accusative / nominative of stem <i>vana</i> present participle → masculine singular locative of root <i>gam</i> (of class 1)
3	<i>rāmaḥ</i> <i>vanam</i> <i>gacchati</i>	verb → present active first person plural of root <i>rā</i> (of class 2) noun → neuter singular accusative / nominative of stem <i>vana</i> verb → present active third person singular of root <i>gam</i> (of class 1)
4	<i>rāmaḥ</i> <i>vanam</i> <i>gacchati</i>	verb → present active first person plural of root <i>rā</i> (of class 2) noun → neuter singular accusative / nominative of stem <i>vana</i> present participle → masculine singular locative of root <i>gam</i> (of class 1)

and represent grammatical classes in Sanskrit. These phases are dictated by the grammatical constraints the language imposes during word formation. For example, the structure of a sentence having words with different phases showing their sequences is shown in figure 3.

Fig. 3 Structure of Phases in SH (image taken from [Sanskrit Heritage Platform's Manual](#))

Among these four solutions (table 13), the first one is more probable and the remaining three are almost rare or may be possible when embedded in a larger context. With these two scenarios of disambiguation and the available data, one can think of two ranking algorithms:

1. use the word frequency: this will be useful to identify the correct segments as in the example *śvetodhāvati* shown earlier in (a)

2. use the stem-morphological analysis frequency: since SH clubs some of the morphological analyses into a single phase as seen in the example *rāmovanaṅgacchati* shown earlier in (b), deciding the frequency of an analysis is not trivial when the phase corresponds to multiple morphological analyses. In such cases, the highest frequency of the morph analyses in that phase is chosen for computational purpose.

Thus we now have two metrics: one with the word frequency and the other with the highest frequency of the morphological analyses of a given word form. The formulae for obtaining the joint probability using word frequency was mentioned in 7.1.1. The corresponding formulae to obtain the joint probability with the stem-morphological analysis frequency is:

$$C_{total} = \prod_{i=1}^n P_{s_i} \quad (4)$$

where, P_{s_i} is the unigram probability of the tuple s_i which represents: [stem, inflectional morphological analysis, base-stem, derivational morphological analysis], and the tuple with the highest frequency is chosen.

8 Observations

8.1 Performance of Ranking in SH

The performance of SH-Ranking using the frequencies from Alignments 1, 2 and 3 over three metrics (word-based, word-transition-based and stem-morph-based) is discussed here. The initial observations are based on the *Bhagavad Gītā* (Vyāsa, 2007) corpus of 700 verses, which is available in both the segmented and unsegmented forms.

Ranking was limited to the first 100 ranks when the word and word-transition metrics were used, and to the first 200 ranks when the stem-morph metrics was used. SH considers the differences in morphological analysis while generating the segmentation solutions. In the word and word-transition metrics, all the solutions having the same word-forms but different morphological analyses were collapsed together, ignoring the morphological analyses and orienting the whole outcome with segmentation-only results. While using the frequencies of stem and morphological analysis, these identical segmentations which differ only in morphological analyses were also collapsed to have an equivalent comparison with the other metrics, even though it would mean losing their main criteria for ranking. In order to account for this, the ranking was limited to a larger window (200) to accept identical segmentations differing only in morphological analyses but having ranks beyond the 100th position.

In the first run of the 700 verses, with the word-transition metrics (reported in Krishnan et al (2023)), it was observed that the correct segmentation was found within the first 100 ranks for 345 / 700 verses, 94 verses had unrecognized chunks and for the remaining verses, the correct segmentation was either ranked beyond the 100th position or not produced at all. For handling the unrecognized chunks, SH lexicon was updated to recognize them. For the

remaining verses it was observed that while SH produced compositional analyses of compounds, the gold segmentation had non-compositional analyses. Hence, the gold segmentations were augmented with compositional analyses of compounds.

With the updated gold segmentations, the segmented solutions of SH were then ranked following the three different metrics from the results of all the three alignments. The relative performance of the three metrics is presented in table 14.

Evaluation: Two strategies were considered for evaluating the segmentation results in all these experiments. The first metrics has a stricter comparison where the evaluation is done based on the entire sentence. The observations shown in table 14 follow this sentence-based evaluation metrics. The second evaluation metrics is described along with the results in section 8.5.²⁸

Table 14 Position Distribution of Bhagavad Gita verses on SH-Ranking

Position	A1 ¹			A2 ²			A3 ³		
	w*t ⁴	w ⁵	m ⁶	w*t	w	m	w*t	w	m
1	399	514	495	415	543	<u>521</u>	414	543	519
2	77	64	63	77	54	54	75	60	56
3	47	36	25	46	24	24	48	18	21
4	27	16	23	31	17	17	27	16	15
5	16	16	14	16	10	11	15	13	13
> 5	101	25	27	83	25	24	87	21	21
Total	667	<u>671</u>	647	668	673	651	666	<u>671</u>	645
Missed	25	21	45	24	19	41	26	21	47
No solution	8	8	8	8	8	8	8	8	8

The best performing metrics are in bold and the second best are underlined.

¹A1 - Alignment 1

²A2 - Alignment 2

³A3 - Alignment 3

¹w*t: word-transition metrics

²w: word metrics

³m: stem-morph metrics

In all the three alignments, the word-metrics performs the best with the correct solution in the first rank for 73.42% of the verses with alignment 1

²⁸The code and results of the evaluation are made available here: [sanskrit-segmentation-evaluation](#)

dataset, 77.57% with alignment 2 and 3 datasets. The percentage of the number of verses which had the correct solution in the top five ranks for the alignments 1, 2 and 3 with the word metrics are respectively, 92.28%, 92.57% and 92.85%. Also, ignoring the compositionality of compounds resulted in an increase in percentage of correct solutions by 46.25% (353 to 673), and correct solutions in first rank by 40.17% (265 to 543, refer word-metrics of Alignment 2 in table 14).

When the stem-morph frequencies are used, there is a minor decrease in the correct segmentation at the first position (when compared to the word metrics). Despite collapsing the identical segmentations together, by ignoring the differences due to morphological analyses, the performance of the stem-morph frequencies in the first five ranks for the three alignments are comparatively lesser (88.57%, 89.57% and 89.14%, respectively).

These observations show that for the word segmentation task, the stem and morphological information are in the first place required for validating the segments, but for ranking these do not contribute much in comparison with the words. The stem-morph frequencies are best used when considering the task of morphological parsing rather than word segmentation alone.

8.2 Performance of rcNN model on *Bhagavad Gītā*

Hellwig and Nehrdich (2018) proposed a character level recurrent and convolutional neural network model (rcNN) for segmentation and used DCS dataset for training. A comparison of this model with SH-Ranking is carried out to understand the performance differences and other intricacies in the two models. Since rcNN produces only one solution while SH-Ranking produces the top n solutions, only the first solution from SH is considered for comparison.

The rcNN model recorded a correct solution for 217 of the 700 verses of *Bhagavad Gītā*. Since rcNN does not handle the terminal sandhis uniformly, these results were passed on to post-processing where terminal sandhis and homonasal consonants were modified all according to the format of the ground truth analysis. Also, rcNN is trained with DCS which has non-compositional analysis for the compounds, according to the context, while SH's compounds have compositional analysis predominantly. Hence, both non-compositional and compositional analyses were considered as correct segmentations during the evaluation. These changes increased the count to 403. In the case of SH-Ranking, it was 543 (alignment 3) with the word-metrics and 521 (alignment 2) with the stem-morph-metrics.

8.3 *Meghadūta*, *Saṁkṣepa-Rāmāyaṇa* and Test Set of rcNN

Meghadūta (Kālidāsa, 1934), is written in two parts containing 121 verses together. Each of the verses was divided into two halves (hemistichs) to facilitate a smooth run of SH-Ranking as most of these verses have large compounds and SH-Ranking fails to produce the results for input with large compounds.

12 hemistichs had chunks unrecognized by SH and for the remaining 230 hemistichs, SH-Ranking (with the word and stem-morph metrics from all the three alignments) and rcNN were tested. SH-Ranking (with Alignment 1 frequencies) produced the correct segmentations for 203 / 230 hemistichs, of which 180 had the correct segmentation in the first rank. On the other hand, rcNN was able to identify the correct segmentation for 81 hemistichs. Expanding the gold segmentations by ignoring the compositionality of certain compounds (depending on the context) resulted into correct segmentation by SH-Ranking for 210 hemistichs and 185 of them in first position. And rcNN was able to predict the correct segmentations for 92 hemistichs. There are about 580 compounds in *Meghadūta* with possibly around 890 binary combinations. Some of these compounds require compositional analyses according to the context where rcNN predicts them as non-compositional. This could be a major reason for the huge difference in the performances of SH-Ranking and rcNN. Also, recently *Meghadūta* corpus had been used as a development corpus to update SH's lexicon and its performance. This could also be contributing to this huge difference.

Śaṅkṣepa-Rāmāyaṇa (Vālmīki, 2002) is an abridged version of the epic *Rāmāyaṇa*, written by Vālmīki, describing the original story in just 100 verses. 18 verses had chunks unrecognized by SH and hence the comparison was done for the remaining 82 verses. The performances of SH-Ranking (with the frequencies from the three alignments) and rcNN are almost similar (39 and 36 verses respectively) when we consider the first position alone. Ignoring the differences due to the compositionality in the gold segmentations increased the performance of SH-Ranking by a mere 4 verses while rcNN was able to predict 20 more verses (56). Although, SH-Ranking produced the correct segmentations for 55 verses across all positions.

The test set of Hellwig and Nehrdich (2018) contains over 21,000 sentences from DCS. Ignoring those sentences which are not recognized by SH, the remaining sentences (18,404) were run on SH-Ranking and rcNN. rcNN performs better than SH with a difference of about 30%. The main reason for this under-performance of SH-Ranking owes to the compositionality of compounds. The gold segmentations were obtained from DCS and they contain non-compositional analyses when the context requires it while SH predominantly produces compositional analyses. Since it is tedious to work on each of these sentences and modify the gold segmentations to accept both compositional and non-compositional analyses, a sample set of the test sentences (597) was taken into consideration. In the initial pass, SH-Ranking (word-metrics of Alignment 1) was able to identify 300 / 597 sentences correctly of which 272 were in the first position. The rcNN model was able to identify the correct segmentation for 367 verses. The gold segmentations were then modified to accept both compositional and non-compositional analyses. Ignoring the differences due to the compositionality, thus, increased the performance of SH-Ranking from 300 to 484 of which 424 were in first position, and the performance of rcNN from 367 to 402.

The performance comparison of SH-Ranking (with the word and stem-morph metrics from all the three alignments) and rcNN on the verses from *Meghadūta* and *Saṅkṣepa-Rāmāyaṇa* (Vālmīki, 2002), and the test sentences from Hellwig and Nehrdich (2018) are recorded in table 15.²⁹

Table 15 Sentence-level performance comparison of *Bhagavad Gītā*, *Meghadūta*, *Saṅkṣepa-Rāmāyaṇa* and Test Sentences from rcNN on SH-Ranking and rcNN models

		SH-Ranking								rcNN
		A1 ¹		A2 ²		A3 ³		A1H ⁴		
		m ⁵	w ⁶	m	w	m	w	m	w	
BG ⁷	T ¹²	692	692	692	692	692	692	692	692	692
	O ¹³	647	671	651	673	645	<u>671</u>	650	671	403
	F ¹⁴	495	514	521	543	519	<u>543</u>	476	490	403
MD ⁸	T	230	230	230	230	230	230	230	230	230
	O	<u>204</u>	210	204	209	204	209	201	209	92
	F	<u>181</u>	185	164	171	168	173	164	164	92
SR ⁹	T	82	82	82	82	82	82	82	82	82
	O	55	55	55	<u>55</u>	55	55	55	55	36
	F	42	42	41	<u>42</u>	40	40	43	43	36
r(a) ¹⁰	T	18,404	18,404	18,404	18,404	18,404	18,404	18,404	18,404	18,404
	O	10,064	10,078	10,063	10,079	10,064	10,078	10,072	<u>10,089</u>	15,184
	F	8,770	8,842	8,971	9,109	9,033	<u>9,162</u>	8,668	<u>8,754</u>	15,184
r(s) ¹¹	T	597	597	597	597	597	597	597	597	597
	O	483	484	483	<u>484</u>	483	<u>484</u>	483	484	402
	F	412	424	409	<u>420</u>	410	<u>420</u>	407	418	402

The best performing metrics / model are in bold and the second best are underlined.

¹A1 - Alignment 1

²A2 - Alignment 2

³A3 - Alignment 3

⁴A1H - Alignment with Hackathon dataset

⁵m - morph-metrics

⁶w - word-metrics

⁷BG - *Bhagavad Gītā*

⁸MD - *Meghadūta*

⁹SR - *Saṅkṣepa-Rāmāyaṇa*

¹⁰r(a) - rcNN test (all)

¹¹r(s) - rcNN test (sample)

¹²T - test sentences

¹³O - solutions obtained

¹⁴F - solutions in first position

Thus, the differences in the way compounds are analysed in rcNN and SH, and also in the DCS dataset, are the main reasons for this performance

²⁹The results after ignoring the compositionality of compounds are recorded in the table.

difference. And that is the major obstacle to solve while trying to align DCS and SH to produce a proper dataset for segmentation tasks in Sanskrit.

8.4 Comparison of SH-Ranking and rcNN with TransLIST

Sandhan et al (2022) provided a performance comparison of rcNN and TransLIST along with various other models using two datasets: SIGHUM and Hackathon. It was observed that with both the datasets TransLIST outperformed the remaining models. SH-Ranking is now tested over the Hackathon dataset.

The Hackathon dataset contains 90,000, 10,332 and 9,963 sentences as train, dev and test set, respectively. The 90,000 sentences from the parallel corpus of the training set were used to create the frequencies for the ranking. 9,910 sentences from the test set were considered for the performance comparison, and the sentence-based comparison is recorded in table 16. Among the SH-Ranking metrics, the morph metrics performed the best with producing the correct solution for 8,080 sentences. But the TransLIST edged the SH-Ranking (morph) by 0.20%.

Table 16 Sentence-level performance comparison of SH-Ranking, rcNN and TransLIST on the Test Set of Hackathon Dataset

	SH-Ranking ¹		rcNN	TransLIST
	m ²	w ³		
Total	9,910	9,910	9,910	9,910
Obtained	<u>8,987</u>	8,989	7,963	8,100
First	<u>8,080</u>	8,055	7,963	8,100

The best performing metrics / model are in bold and the second best are underlined.

¹Alignment with Hackathon dataset

²m - morph-metrics

³w - word-metrics

8.5 Word-level evaluation

A word-level evaluation is carried out where the macro-averaged precision, recall and F-score were taken into consideration. This involves a counter-based evaluation strategy where the counter of each of the ground truth segments (words) of a solution are compared with the counter of each of the predicted segments. And then the macro-averaged precision, recall, etc. over all the solutions are computed. Table 17 gives a summary of the evaluation results of SH-Ranking (with the word-based and stem-morph-based metrics) and rcNN over *Bhagavad Gītā*, *Meghadūta*, *Saṅkṣepa-Rāmāyaṇa* and Test Sentences from

rcNN. And table 18 gives the comparison of SH-Ranking, rcNN and TransLIST over the Hackathon Test set.³⁰

Table 17 Word-level evaluation of SH-Ranking and rcNN models on *Bhagavad Gītā*, *Meghadūta*, *Sanikṣepa-Rāmāyaṇa* and Test Sentences from rcNN

		SH-Ranking								rcNN
		A1 ¹		A2 ²		A3 ³		A1H ⁴		
		m ⁵	w ⁶	m	w	m	w	m	w	
BG ⁷	P ¹²	97.75	97.79	98.08	<u>98.14</u>	98.10	98.16	97.35	97.37	95.39
	R ¹³	97.17	97.25	97.54	<u>97.64</u>	97.59	97.66	96.69	96.78	94.56
	F ¹⁴	97.45	97.51	98.80	<u>97.87</u>	97.83	97.89	97.00	97.06	94.92
MD ⁸	P	97.80	<u>97.74</u>	96.71	97.07	96.82	97.12	96.95	96.52	91.93
	R	97.98	<u>97.85</u>	97.14	97.34	97.23	97.39	97.14	97.76	90.90
	F	97.87	<u>97.78</u>	96.90	97.19	97.00	97.24	97.02	96.62	91.35
SR ⁹	P	<u>93.11</u>	92.20	92.45	92.14	92.56	91.57	93.19	91.45	92.85
	R	93.49	92.84	93.03	92.84	93.13	92.32	<u>93.33</u>	92.94	91.26
	F	93.25	92.46	92.69	92.43	92.79	91.89	<u>93.21</u>	92.64	91.98
r(a) ¹⁰	P	85.28	85.34	85.73	85.93	85.78	<u>85.96</u>	85.14	85.24	96.09
	R	87.83	87.97	88.34	88.58	88.40	<u>88.64</u>	87.62	87.80	96.37
	F	86.31	86.41	86.80	87.02	86.85	<u>87.06</u>	86.14	86.28	96.17
r(s) ¹¹	P	91.90	92.14	91.93	<u>92.24</u>	91.95	92.21	91.83	92.06	93.86
	R	93.45	93.71	93.51	<u>93.80</u>	93.52	93.78	93.38	93.60	94.61
	F	92.59	92.84	92.63	<u>92.93</u>	92.65	92.91	92.52	92.75	94.10

The best performing metrics / model are in bold and the second best are underlined.

¹A1 - Alignment 1

²A2 - Alignment 2

³A3 - Alignment 3

⁴A1H - Alignment with Hackathon dataset

⁵m - morph-metrics

⁶w - word-metrics

⁷BG - *Bhagavad Gītā*

⁸MD - *Meghadūta*

⁹SR - *Sanikṣepa-Rāmāyaṇa*

¹⁰r(a) - rcNN test (all)

¹¹r(s) - rcNN test (sample)

¹²P - Precision

¹³R - Recall

¹⁴F - F-Score

Considering the sentence-level evaluation, SH-Ranking with alignment frequencies performed better for *Bhagavad Gītā*, *Meghadūta*, *Sanikṣepa-Rāmāyaṇa* and the sample set of rcNN-test where compositional analyses of compounds

³⁰These tables record the results after ignoring the compositionality of compounds.

Table 18 Word-level evaluation of SH-Ranking, rcNN and TransLIST models on the Hackathon dataset

	SH-Ranking ¹		rcNN	TransLIST
	m ²	w ³		
P ⁴	97.48	<u>97.43</u>	97.35	97.31
R ⁵	<u>96.95</u>	96.91	97.67	96.90
F ⁶	<u>97.19</u>	97.14	97.40	97.09

The best performing metrics / model are in bold and the second best are underlined.

¹Alignment with Hackathon dataset

²m - morph-metrics

³w - word-metrics

⁴P - Precision

⁵R - Recall

⁶F - F-Score

were accepted in the gold segmentations. The performances of the three alignment frequencies are almost similar, except for *Meghadūta* where the alignment 1 frequencies outperformed the rest, and for *Bhagavad Gītā*, alignment 2 and 3 frequencies outperformed the rest. The rcNN model outperformed the rest with a huge margin in the case of rcNN-test sentences. With the Hackathon test set, TransLIST edged over SH-Ranking by a margin of 0.20% and performed better than rcNN by a margin of 1.38%.

Considering the word-level evaluation, alignment 3 frequencies performed the best for *Bhagavad Gītā*, alignment 1 frequencies for *Meghadūta*, while the rcNN model and alignment 2 frequencies performed equally well for *San̐kṣepa-Rāmāyaṇa*. For the rcNN-test sentences, the rcNN model outperformed the remaining with a huge margin mainly due to the difference in compositional analysis of compounds. For the sample set considered from rcNN-test, accepting compositional analyses, the rcNN model still performed better followed by the alignment 2 frequency with a difference of less than 1.5%. With the Hackathon test set, the morph-based SH-Ranking edged over rcNN and TransLIST in terms of precision while rcNN had better recall and F-score values.

Upon comparing the performances of the four alignment datasets, two important observations emerge. From the sentence-level evaluation, it was noticed that the average difference between the maximum and minimum values of correct segmentations in the first position, calculated across the test sets, was 5.56%. This finding highlights the significant impact of having more data, as it effectively pushes up the correct solutions towards the top rank.

However, when considering the precision values from the word-based evaluation across the alignment datasets, the average difference (percentage) between the maximum and minimum values was only 0.99%. Similarly, for

Recall and F-Score, the average differences between the maximum and minimum values were 0.96% and 0.97%, respectively. These results suggest that while expanding the datasets has not led to a substantial improvement in performance, the primary objective of elevating the correct solutions to higher rankings has been achieved.

9 Inferences and Discussion

The alignment process and the experiments with the datasets obtained thereafter provided us a few important cues to proceed forward towards the development of SH. Several anomalies of SH were discovered and corrected with the help of the results from the experiments. The alignment process also helped addressing the limitations of the reference corpus, validating and normalizing it before generating the unified dataset. SH is further enhanced with a ranking mechanism that leverages the unified datasets. The datasets were also shown to be of greater use to the task of Word Segmentation. What follows is a brief account of all the inferences from these aforementioned developments.³¹

9.1 Unified Dataset Constructed via alignment of DCS and SH

The unified dataset used for the ranking mechanism is specifically built for the tasks of Word Segmentation and Morphological Parsing and proved to be of greater impact when tested on different texts. The comparison with the neural network models also revealed similar performances of SH justifying that n-gram models work on par and sometimes better than NN models for Sanskrit Word Segmentation. This unified dataset comes with most of the limitations of DCS.

First, sentence boundaries are unclear with respect to DCS. In some cases, the hemistichs of the same verse are considered as two different sentences, while huge sentences are also found in the list of DCS sentences. This restricts the dataset to be not used for tasks like dependency analysis, although dedicated fields have been provided for the universal POS tags, language specific POS tags and dependency relations, etc. Second, chunk boundaries are also not uniformly set and a comparison with the chunk boundaries (section 5.1) of SH shows the impact of this limitation.

But with more analysis on the remaining sentences of DCS and with SH's lexicon getting updated with unrecognized stems and possible named entities, more sentences can be aligned to extend this dataset.

9.2 Improvement to the Heritage Segmenter

Non-determinism in *sandhi* resolution: First, SH has been extended to prioritize the list of possible segmentation solutions by keeping a bucket of n

³¹The details of the DCS-SH alignment (code, data and references) are released publicly and can be accessed here: https://github.com/SriramKrishnan8/dcs_sh_alignment

solutions, where n can be an arbitrary number depending upon the requirement of the user. SH uses a Finite-State-Automata based approach for the segmentation, producing all possible segmentations due to the non-deterministic nature of Sanskrit Word Segmentation. And the ranking is integrated as a dovetailing mechanism on top of this segmentation. The current ranking algorithm uses a joint probability of the unigram frequencies of words or morphological analyses.

Second, the dovetailing algorithm helps in truncating the entire list of possible solutions to a limited set without much significant loss of recall. The frequencies used for ranking based on different parameters (word, transition, stem-morph) showed that with a joint probability the precision achieved results on par with the neural network model for sentences recognized by SH. Also, to make it consistent with the data structures used in SH, these frequencies were integrated into a decorated trie structure holding the keys (word / stem-morph tuple) and frequencies, which further helped in speeding up the access rate of the word-frequency pairs. One main advantage of this structure is in its sharing of suffixes, very similar to the trie structures built for the lexicon.

Third, the probabilities of stem and morphological analysis showed not much improvement for the word segmentation task indicating that these are best used with the morphological parsing task. Although, word segmentation requires the results to have segmented and morphologically and lexically valid word forms and with non-determinism at the level of morphological analysis too (due to homonymy and syncretism), these frequencies would be of greater use in narrowing down to the best morphological analyses of the best segments chosen. Thus, an integrated approach of using word probabilities to rank the solutions based on word forms and then using the stem and morph probabilities to rank the morph possibilities among each of the segmentations could perform better.

Fourth, comparisons with other segmenters showed that SH has two major advantages over other systems. One, distinguishing the *sandhi* between words with the *sandhi* between components of compounds. This identification of compound form is crucial in the tasks of compound analysis and dependency parsing. Two, producing the morphologically and lexically valid splits. Ranking the solutions might result into an unintended segmentation to be at the top, but the overall production of possible segmentations never goes wrong. Although, out-of-vocabulary words and non-compositional compounds are to be handled effectively.

Fifth, the alignment process provides for every word in the sentence, its base, derivational morphological analysis, derived stem, inflectional morphological analysis, preverb, phase details along with all the other default information obtained from DCS.

Finally, the current work focused on a ranking mechanism over an integrated approach of word segmentation and morphological parsing using the Heritage tools. Although the alignment processes have proven effective in pushing the correct solutions towards the top ranking, they have not been as helpful

in identifying segments, as observed in the word-level evaluation. The compositionality of compounds posed a major obstacle, making the non-determinism in the *Sandhi* resolution within a compound more difficult to handle than the non-determinism in the *Sandhi* resolution between words. It is thus crucial to distinguish the *Sandhi* between the compounds and the *Sandhi* between words as *Sandhi* resolution in a compound requires the compositionality of the compound according to the context in addition to the *Sandhi* rules and lexical and morphological information. Therefore, it is essential to direct further attention towards determining the compositionality of compounds and exploring how it can contribute to word segmentation.

9.3 Integrating the updates with *Samśādhani*³² tools

The *Samśādhani*³² platform provides various tools for processing Sanskrit texts of which the *Sandhi-Vicchedikā* (*Sandhi* Splitter) and *Anusārakam*³³ make use of the segmented solutions from SH. Since SH-Ranking produces best possible segmentations, these are fed to the *Sandhi-Vicchedikā* and *Anusārakam* via a pipeline established as a collaborative effort between these two systems to enable cross-platform analysis of sentences (Huet and Kulkarni, 2014). The best segmentation solution, along with the best possible morphological analysis for each of the segments, is fed to the *Anusārakam* (interface of *Samśādhani*) for further process of Dependency Parsing on the segmented sentence. Thus, the overall ranking mechanism using an expanded dataset would result in pushing the correct solutions towards the top, and in effect, helping these tools by providing the preliminary information viz. the most probable segmented forms and morphological analysis.

9.4 Improvements to the reference corpora

A detailed analysis on DCS' inconsistencies of analyses, errors due to chunk, segmentation and compound splits and mismatches due to the design of DCS showed the major limitations of DCS, especially in such a task of aligning its results with another platform like SH. In this process, a mapper of the morphological constructions of SH with the morphological tags of DCS was generated which was collaborated with two other existing tagsets (in SCL and SLP³⁴). This was used in the alignment process and also finds its use during the cross platform analysis of sentences. With respect to representation of stems, the resultant unified dataset made sure that both the derived and base forms are annotated rather than just one of these or any other intermediate construction. Mapping the senses of the stems are out of scope in the current research, but could be of greater importance to handle the sentences which have multiple alignments.

³²www.sanskrit.uohyd.ac.in/scl

³³A machine translation and accessor system from Sanskrit to other languages like Hindi, Marathi and Telugu.

³⁴<https://sanskritlibrary.org/helpmorphids.html>

The intention behind this work was to provide a dataset which is rich in morphological and lexical analyses, and which provides as much details as possible regarding the sentences and words. The alignment process did pave the way to building such a dataset which was used on the existing Segmenter to improve it further. Thus, the resultant dataset could give rise to a homogeneous gold corpus which could in turn be used for various subsequent tasks of sentential analysis.

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